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Department : B.TECH(AIDS)

Subject : Problem Solving Using Python

Batch : 03

PROJECT: Student Record System using File Handling

AIM

To develop a Python program that performs read and write operations on a file to store, retrieve, and manage student records.

ALGORITHM

1. Start the program
2. Create a file (students.txt) if it doesn't exist
3. Display menu:
    - Add Record
    - View Records
    - Search Record
    - Exit
4. If user selects Add:
    - Input name, roll number, marks
    - Write data to file
5. If user selects View:
    - Read and display all records
6. If user selects Search:
    - Input name

- Search and display matching record

7. Repeat until user exits
8. End

CODE

```
def add_record():
    with open("students.txt", "a") as file:
        name = input("Enter Name: ")
        roll = input("Enter Roll No: ")
        marks = input("Enter Marks: ")

        file.write(f"{name},{roll},{marks}\n")
        print("✅ Record Added Successfully!")

def view_records():
    try:
        with open("students.txt", "r") as file:
            data = file.readlines()
            if not data:
                print("⚠️ No records found!")
                return

            print("\n📋 Student Records:")
            for line in data:
                name, roll, marks = line.strip().split(",")
                print(f"Name: {name}, Roll: {roll}, Marks: {marks}")

    except FileNotFoundError:
        print("⚠️ File not found!")

def search_record():
    name_search = input("Enter name to search: ").lower()

    try:
        with open("students.txt", "r") as file:
            found = False
            for line in file:
                name, roll, marks = line.strip().split(",")
```

```
            if name.lower() == name_search:
                print(f"🔍 Found → Name: {name}, Roll: {roll}, Marks: {marks}")
                found = True

        if not found:
            print("❌ Record not found!")

    except FileNotFoundError:
        print("⚠️ File not found!")

def main():
    while True:
        print("\n📁 STUDENT RECORD SYSTEM")
        print("1. Add Record")
        print("2. View Records")
        print("3. Search Record")
        print("4. Exit")

        choice = input("Enter choice: ")

        if choice == "1":
            add_record()
        elif choice == "2":
            view_records()
        elif choice == "3":
            search_record()
        elif choice == "4":
            print("👋 Exiting program...")
            break
        else:
            print("⚠️ Invalid choice!")

main()
```

🖥️ SAMPLE OUTPUT

```
📁 STUDENT RECORD SYSTEM
1. Add Record
2. View Records
3. Search Record
4. Exit

Enter choice: 1
Enter Name: Gokul
Enter Roll No: 101
Enter Marks: 89
✅ Record Added Successfully!

Enter choice: 2

📋 Student Records:
Name: Gokul, Roll: 101, Marks: 89
```

📊 RESULT

The Student Record System was successfully implemented using Python file handling. The program allows storing, retrieving, and searching student data efficiently using file operations.